Probability & Statistics

Unit-IV

Test of Hypothesis

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To reach decisions about populations on the basis of sample information, we make certain assumptions about the populations involved. Such assumptions, which may or may not be true, are called statistical hypothesis. By testing a hypothesis is meant a process for deciding whether to accept or reject the hypothesis.

Null Hypothesis

The hypothesis formulated for the sake of rejecting it, under the assumption that it is true, is called the null hypothesis and is denoted by H_0. To test whether one procedure is better than another, we assume that there is no difference between the procedures. Similarly to test whether there is a relationship between two variates, we take H_0 that there is no relationship. By accepting a null hypothesis, we mean that on the basis of the statistic calculated from the sample, we do not reject the hypothesis.

Alternative Hypothesis

Any hypothesis which contradicts the Null Hypothesis is called an Alternative Hypothesis, usually denoted by H_1. The two hypotheses H_0 and H_1 are such that if one is true, the other is false.

For example, if we want to test the null hypothesis that the population has a specified mean μ_0 (say) i.e.,

Null hypothesis H_0: $\mu = \mu_0$

Then alternative hypothesis would be

Alternative hypothesis H_1: (i) $\mu \neq \mu_0$ (either $\mu > \mu_0$ or $\mu < \mu_0$) or

(ii) $\mu > \mu_0$ or

(iii) $\mu < \mu_0$

The alternative hypothesis (i) is known as a two-tailed alternative and (ii) is known as right-tailed alternative and (iii) is known as left-tailed alternative

Errors

Type I Error

If a hypothesis (H_0) is rejected while it should have been accepted, we say that a Type I error has been committed. It is also called α error.

If we write P(reject H_0 when it is true) = P(Type I error) = α

Then α is called size of type I error and is known as producer's risk.

Type II error

If a hypothesis (H_0) is accepted while it should have been rejected, we say that a Type II error has been made. It is also called β error.

If we write P(accept H_0 when it is wrong) = P(Type II error) = β

Then β is called size of type II error and is known as consumer's risk.

The statistical testing of hypothesis aims at limiting the Type I error to a preassigned value (say: 5% or 1%) and to minimize the Type II error.

Table of error types	Null hypothesis (H_0) is

		True	False
Decision about null hypothesis (H_0)	**Not reject (Accept)**	Correct inference	Type II error
	Reject	Type I error	Correct inference

Level of Significance (α)

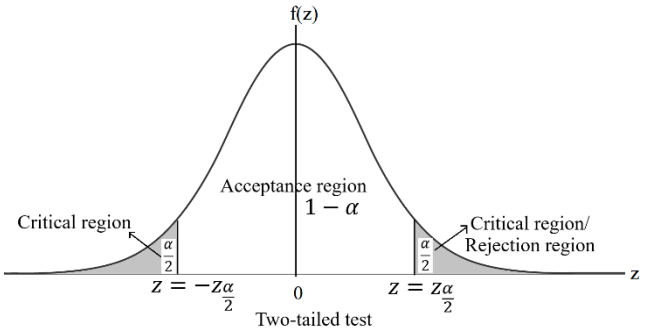
The probability level below which we reject the hypothesis is known as the level of significance. The level of significance denoted by α is the confidence with which we reject or accept the null hypothesis H_0 i.e., it is the maximum possible probability with which we are willing to risk an error in rejecting H_0 when it is true. The level of significance is generally specified before a test procedure so that the results obtained may not influence our decision. In practice, we take either 5% (i.e., 0.05) or 1% (i.e., 0.01) level of significance.

The region in which a sample value falling is rejected, is known as critical region. The critical region depends on the test statistic and the type of alternative hypothesis and α. Thus, the probability of the value of the variate falling in the critical region is the level of significance.

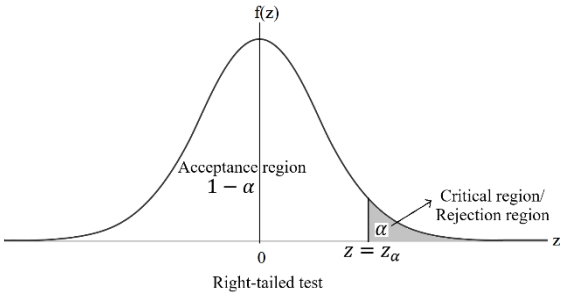
One-tailed and two-tailed tests

Depending on the nature of the problem, we use a single-tailed test or double-tailed test to estimate the significance of a result.

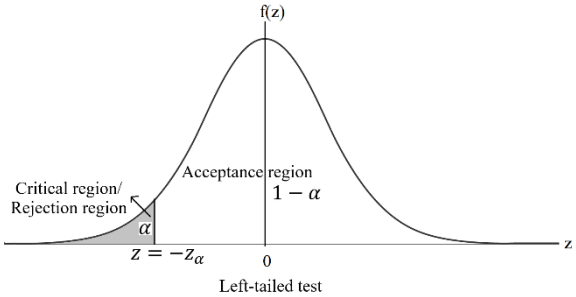
If the alternative hypothesis is of the type (i) $H_1: \mu \neq \mu_0$, then two-tailed test is applied.
In a two-tailed test, the areas of both the tails of curve representing the sampling distribution are taken into account.



If the alternative hypothesis is of the type (ii) $H_1: \mu > \mu_0$ or (iii) $H_1: \mu < \mu_0$ then one-tailed test is applied.
In a one-tailed test, only the area on one side of an ordinate is taken into consideration.
If $H_1: \mu > \mu_0$ then a right tailed test is applied.



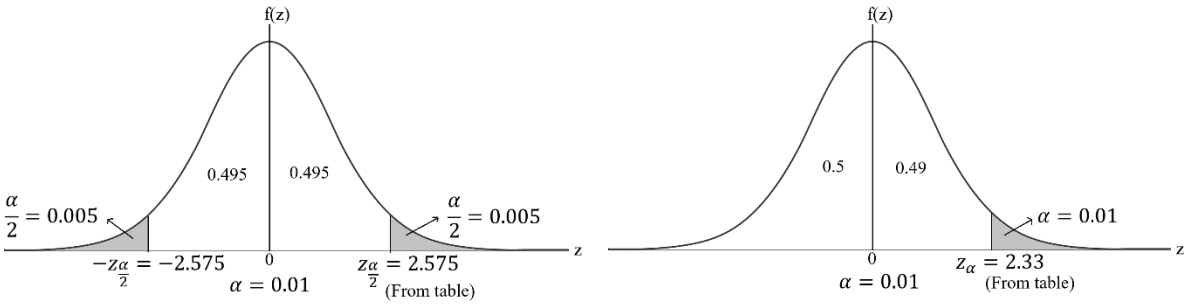
If $H_1: \mu < \mu_0$ then a left tailed test is applied.



Critical values (z_α) of z

Level of significance (α)	**Critical values of z**		
	1% ($\alpha = 0.01$)	**5% ($\alpha = 0.05$)**	**10% ($\alpha = 0.10$)**

Two-tailed test	$\left\|z_{\frac{\alpha}{2}}\right\| = 2.575$	$\left\|z_{\frac{\alpha}{2}}\right\| = 1.96$	$\left\|z_{\frac{\alpha}{2}}\right\| = 1.645$
Right tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$



The value of the test statistic which separates the rejection region (critical region) and the acceptance region is called the critical value or significant value. This value is dependent on

(i) the level of significance used, and

(ii) the alternative hypothesis, whether it is one-tailed or two-tailed.

Tests of significance

The procedure which enables us to decide whether to accept or reject the hypothesis is called the test of significance.

Procedure for testing of hypothesis

Step 1. Null Hypothesis (H_0): Define a null hypothesis taking into consideration the nature of the problem and data involved.

Step 2. Alternative Hypothesis (H_1): Set up the alternative hypothesis so that we could decide whether we should use one-tailed or two-tailed test.

Step 3. Level of Significance (α): Select the appropriate level of significance depending on the reliability of the estimates and permissible risk. If it is not given, then we choose 5% level of significance.

Step 4. Test Statistic: Compute the test statistic $z = \frac{t - E(t)}{S.E.(t)}$ under the null hypothesis. Here t is a sample statistic and S.E. is the standard error of t.

Step 5. Conclusion: We compare the computed value of the test statistic z with the critical value z_α at given level of significance.

Test of significance of a single mean (Large samples, $n \geq 30$)

Let a random sample of size n ($n \geq 30$) has the sample mean $\overline{x}$, and μ be the population mean and σ be the population standard deviation. Also, the population mean has a specified value μ_0.

Step 1. Null Hypothesis (H_0): $\mu = \mu_0$.

Step 2. Alternative Hypothesis (H_1): (i) $\mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) (Two-tailed test)

Or (ii) $\mu > \mu_0$ (Right tailed test)

Or (iii) $\mu < \mu_0$ (Left tailed test)

Step 3. Level of Significance (α): Set the level of significance α.

Critical value of z is z_α (one-tailed test) or $z_{\frac{\alpha}{2}}$ (two-tailed test).

Step 4. Test Statistic: $z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

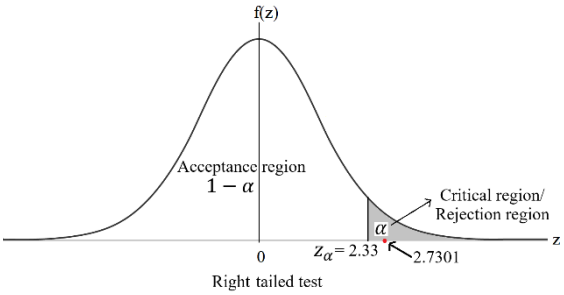
Step 5. Conclusion: If $|z| < |z_\alpha|$, we accept the null hypothesis H_0.
If $|z| > |z_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Note: If the population standard deviation σ is unknown, then use the test statistic $z = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}}$, where s is the sample standard deviation.

Examples

Example 1: According to the norms established for a mechanical aptitude test, persons who are 18 years old have an average height of 73.2 with a standard deviation of 8.6. If 45 randomly selected persons of that age averaged 76.7, test the hypothesis $\mu = 73.2$ against the alternative hypothesis $\mu > 73.2$ at the 0.01 level of significance.

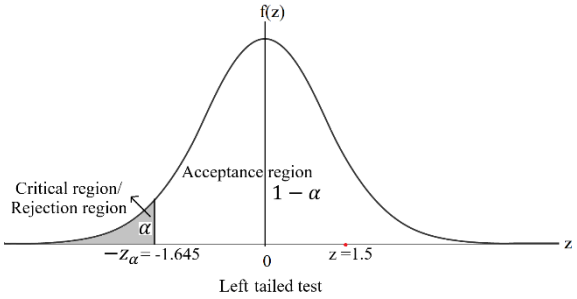
Solution. Given
population mean $\mu = 73.2$,
population standard deviation $\sigma = 8.6$,
sample size $n = 45$ (≥ 30, large sample),
sample mean $\overline{x} = 76.7$
Step 1. Null Hypothesis (H_0): $\mu = 73.2$.
Step 2. Alternative Hypothesis (H_1): $\mu > \mu_0$ (Right tailed test)
Step 3. Level of Significance (α): $\alpha = 0.01$. Critical value of z is $z_\alpha = 2.33$.
Step 4. Test Statistic: $z = \frac{\overline{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow z = \frac{76.7-73.2}{\frac{8.6}{\sqrt{45}}} = 2.7301$
$\therefore |z| > |z_\alpha|$
Step 5. Conclusion: Since $|z| > |z_\alpha|$, the null hypothesis is rejected and alternative hypothesis is accepted i.e., $\overline{x}$ and μ differ significantly.



Example 2: An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the claim at 0.05 level of significance.

Solution. Given
sample size $n = 36$ (≥ 30, large sample)
sample mean $\overline{x} = 11$,
sample variance $s^2 = 16$,
sample standard deviation $s = 4$,
Step 1. Null Hypothesis (H_0): $\mu = 10$.
Step 2. Alternative Hypothesis (H_1): $\mu < 10$ (Claim) (Left tailed test)
Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value of z is $z_\alpha = -1.645$.
Step 4. Test Statistic: $z = \frac{\overline{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow z = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}} \Rightarrow z = \frac{11-10}{\frac{4}{\sqrt{36}}} = -1.5$
$\therefore |z| < |z_\alpha|$

Step 5. Conclusion: Since $|z| < |z_\alpha|$, the null hypothesis is accepted and alternative hypothesis is rejected i.e., $\mu < 10$ is rejected.



Example 3: A sample of 400 items is taken from a population whose standard deviation is 10. Then mean of the sample is 40. Test whether the sample has come from a population with mean 38.

Solution. Given

sample size $n = 400$ (≥ 30, large sample)

sample mean $\overline{x} = 40$,

population mean $\mu = 38$,

population standard deviation $\sigma = 10$

Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $\mu = 38$.

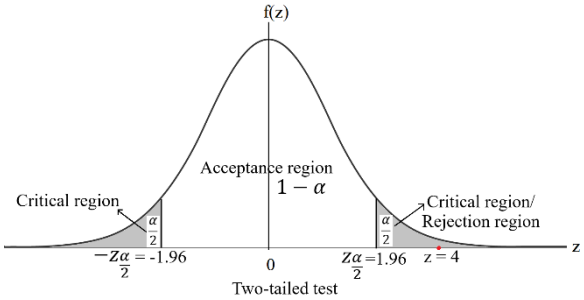
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $\mu \neq 38$ (Two-tailed test)

Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$. Critical value of z is $\left|z_{\frac{\alpha}{2}}\right| = 1.96$.

Step 4. Test Statistic: $z = \frac{\overline{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow z = \frac{40-38}{\frac{10}{\sqrt{400}}} = 4$

$\therefore |z| > \left|z_{\frac{\alpha}{2}}\right|$

Step 5. Conclusion: Since $|z| > \left|z_{\frac{\alpha}{2}}\right|$, the null hypothesis is rejected and alternative hypothesis is accepted i.e., the sample is not from a population whose mean is 38.



Test of significance of two means (Large samples, $n \geq 30$)

(Test of significance for difference of means of two large samples)

Let $\overline{x}_1$ and $\overline{x}_2$ be the sample means of two independent large random samples of sizes n_1 and n_2 drawn from two populations having means μ_1 and μ_2 and standard deviations σ_1 and σ_2. To test whether the two population means are equal or not.

Let a random sample of size n $(n \geq 30)$ has the sample mean $\overline{x}$, and μ be the population mean and σ be the population standard deviation. Also, the population mean has a specified value μ_0.

Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $\mu_1 = \mu_2$.

Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: (i) $\mu_1 \neq \mu_2$ (i.e., $\mu_1 > \mu_2 or \mu_1 < \mu_2$) (Two-tailed test)

Or (ii) $\mu_1 > \mu_2$ (Right tailed test)

Or (iii) $\mu_1 < \mu_2$ (Left tailed test)

Step 3. Level of Significance (α): Set the level of significance α.
Critical value of z is z_α (one-tailed test) or $z_{\frac{\alpha}{2}}$ (two-tailed test).

Step 4. Test Statistic: $z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}$

Step 5. Conclusion: If $|z| < |z_\alpha|$, we accept the null hypothesis H_0.
If $|z| > |z_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Note: If the two samples are drawn from two populations with unknown standard deviations σ_1 and σ_2 (variances ${\sigma_1}^2$ and ${\sigma_2}^2$), then ${\sigma_1}^2$ and ${\sigma_2}^2$ can be replaced by sample variances ${s_1}^2$ and ${s_2}^2$, provided both the samples sizes n_1 and n_2 are large. In this case, the test statistic is $z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}}$.

Examples

Example 1: The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D. 2.5 inches?

Solution. Let μ_1 and μ_2 be the means of the two populations. Given
sample size $n_1 = 1000$ (≥ 30, large sample)
sample size $n_2 = 2000$ (≥ 30, large sample)
sample mean $\overline{x}_1 = 67.5$,
sample mean $\overline{x}_2 = 68$,
population standard deviation (S.D.) $\sigma = 2.5$ $(\sigma_1 = \sigma_2 = \sigma)$

Step 1. Null Hypothesis (H_0): $\mu_1 = \mu_2$ and $\sigma = 2.5$ i.e., the samples have been drawn from the same population of S.D. $\sigma = 2.5$.

Step 2. Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (Two-tailed test)

Step 3. Level of Significance (α): $\alpha = 0.05$ (95% confidence)
Critical value of z is $\left|z_{\frac{\alpha}{2}}\right| = 1.96$.

Step 4. Test Statistic: $z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}} \Rightarrow z = \frac{67.5 - 68}{\sqrt{\frac{2.5^2}{1000} + \frac{2.5^2}{2000}}} = -5.16$

$\therefore |z| = 5.16 \; and \; |z| > \left|z_{\frac{\alpha}{2}}\right|$

Step 5. Conclusion: Since $|z| > \left|z_{\frac{\alpha}{2}}\right|$, the null hypothesis is rejected at 5% level of significance and alternative hypothesis is accepted i.e., the samples are not drawn from the same population of S.D. 2.5 inches.

Example 2: The average marks scored by 32 boys is 72 with a S.D. of 8, while that for 30 girls is 70 with a S.D. of 6. Does this indicate that the boys perform better than girls at level of significance 0.05?

Solution. Let μ_1 and μ_2 be the means (mean marks) of the two populations (boys and girls). Given
sample size $n_1 = 32$ (≥ 30, large sample)
sample size $n_2 = 30$ (≥ 30, large sample)
sample mean $\overline{x}_1 = 72$,
sample standard deviation $s_1 = 8$,

sample mean $\overline{x}_2 = 70$,
sample standard deviation $s_2 = 6$
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $\mu_1 = \mu_2$
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $\mu_1 > \mu_2$ (Right tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$ (95% confidence)
Critical value of z is $|z_\alpha| = 1.645$.
Step 4. Test Statistic: $z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}} \Rightarrow z = \frac{72-70}{\sqrt{\frac{8^2}{32} + \frac{6^2}{36}}} = 1.1547$
$\therefore |z| < |z_\alpha|$
Step 5. Conclusion: Since $|z| < |z_\alpha|$, the null hypothesis is accepted at 5% level of significance and conclude that the performance of boys and girls is the same.

Example 3: A simple sample of the height of 6400 Englishmen has a mean of 67.85 inches and a S.D. of 2.56 inches while a simple sample of heights of 1600 Australians has a mean of 68.55 inches and S.D. 2.52 inches. Do the data indicate that the Australians are on the average taller than the Englishmen? (Use α as 0.01).
Solution. Let Englishmen represent the first population and Australians represent the second population. Then μ_1 is the average height of Englishmen and μ_2 is the average height of Australians.
Given
sample size $n_1 = 6400$ (≥ 30, large sample)
sample size $n_2 = 1600$ (≥ 30, large sample)
sample mean $\overline{x}_1 = 67.85$,
sample standard deviation $s_1 = 2.56$,
sample mean $\overline{x}_2 = 68.55$,
sample standard deviation $s_2 = 2.52$
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $\mu_1 = \mu_2$
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $\mu_1 < \mu_2$ (Left tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.01$ (99% confidence)
Critical value of z is $|z_\alpha| = 2.33$.
Step 4. Test Statistic: $z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}} \Rightarrow z = \frac{67.85-68.55}{\sqrt{\frac{2.56^2}{6400} + \frac{2.52^2}{1600}}} = -9.9$
$\therefore |z| > |z_\alpha|$
Step 5. Conclusion: Since $|z| > |z_\alpha|$, the null hypothesis is reject at 1% level of significance and conclude that the Australians are taller than the Englishmen.

Notation: Population proportion is denoted by P and sample proportion is denoted by p.

Test of significance for single proportion (Large samples, $\boldsymbol{n \geq 30}$)
Suppose a large random sample of size n ($n \geq 30$) has a sample proportion p of members possessing a certain attribute (i.e., proportion of success). To test the hypothesis that the proportion P in the population has a specified value P_0.
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P = P_0$ (P_0 is a particular value of P)
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: (i) $P \neq P_0$ (i.e., $P > P_0$ or $P < P_0$) (Two-tailed test)
Or (ii) $P > P_0$ (Right tailed test)
Or (iii) $P < P_0$ (Left tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: Set the level of significance α.
Critical value of z is z_α (one-tailed test) or $z_{\frac{\alpha}{2}}$ (two-tailed test).

Step 4. Test Statistic: $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$, where p is the sample proportion which is approximately normally distributed and $Q = 1 - P$.

Step 5. Conclusion: If $|z| < |z_\alpha|$, we accept the null hypothesis H_0.
If $|z| > |z_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Examples

Example 1: Experience had shown that 20% of a manufactured product is of the top quality. In one day's production of 400 articles only 50 are of top quality. Test the hypothesis at 0.05 level of significance.

Solution. Given

population proportion $P = \frac{20}{100} = 0.2$, $Q = 1 - P = 1 - 0.2 = 0.8$,

sample size $n = 400$ (> 30, large sample)

sample proportion $p = \frac{50}{400} = 0.125$

Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P = 0.2$

Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $P \neq 0.2$ (Two-tailed test)

Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$ (95% confidence)

Critical value of z is $\left|z_{\frac{\alpha}{2}}\right| = 1.96$.

Step 4. Test Statistic: $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} \Rightarrow z = \frac{0.125-0.2}{\sqrt{\frac{0.2\times0.8}{400}}} = -3.75$

$\therefore |z| = 3.75 \; and \; |z| > \left|z_{\frac{\alpha}{2}}\right|$

Step 5. Conclusion: Since $|z| > \left|z_{\frac{\alpha}{2}}\right|$, the null hypothesis is rejected at 5% level of significance and alternative hypothesis is accepted i.e., $P \neq 0.2$

Example 2: In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers?

Solution. Given

sample size $n = 600$ ($\geq$ 30, large sample)

sample proportion $p = \frac{325}{600} = 0.5417$

population proportion of smokers in the city $P = \frac{1}{2} = 0.5, Q = 1 - P = 1 - 0.5 = 0.5$

Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P = 0.5$

Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $P > 0.5$ (Right tailed test)

Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$ (95% confidence)

Critical value of z is $|z_\alpha| = 1.645$.

Step 4. Test Statistic: $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} \Rightarrow z = \frac{0.5417-0.5}{\sqrt{\frac{0.5\times0.5}{600}}} = 2.04$

$\therefore |z| = 2.04 \; and \; |z| > |z_\alpha|$

Step 5. Conclusion: Since $|z| > |z_\alpha|$, we reject the null hypothesis at 5% level of significance and accept the alternative hypothesis and conclude that the majority of men in the city are smokers.

Example 3: A manufacturer claimed that at least 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipment revealed that 18 were faulty. Test his claim at 5% level of significance?

Solution. Given
population proportion $P = \frac{95}{100} = 0.95, Q = 1 - P = 1 - 0.95 = 0.05$
sample size $n = 200$ (≥ 30, large sample)
Number of pieces conforming to specification = 200-18=182
sample proportion of pieces conforming to specifications is $p = \frac{182}{200} = 0.91$
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P = 0.95$
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $P < 0.95$ (Left tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$ (95% confidence)
Critical value of z is $|z_\alpha| = 1.645$.
Step 4. Test Statistic: $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} \Rightarrow z = \frac{0.91-0.95}{\sqrt{\frac{0.95\times 0.05}{200}}} = -2.59$
$\therefore |z| = 2.59 \; and \; |z| > |z_\alpha|$
Step 5. Conclusion: Since $|z| > |z_\alpha|$, we reject the null hypothesis H_0 at 5% level of significance. We accept the alternative hypothesis H_1 and conclude that the manufacturer's claim is rejected.

Test for equality of two proportions (Large samples, $n \geq 30$)
(Test of significance of difference between two sample proportions)
Let p_1 and p_2 be the sample proportions in two large random samples of sizes n_1 and n_2 drawn from two populations having proportions P_1 and P_2.
To test whether the two samples have been drawn from the same population.
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P_1 = P_2$.
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $P_1 \neq P_2$ (Two-tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: Set the level of significance α.
Critical value of z is $z_{\frac{\alpha}{2}}$.
Step 4. Test Statistic: There are two ways of computing a test statistic z

(a) When the population proportions P_1 and P_2 are known

$$z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}, where \; Q_1 = 1 - P_1, Q_2 = 1 - P_2$$

(b) When the population proportions P_1 and P_2 are not known but the sample proportions p_1 and p_2 are known

$$z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, where \; p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, q = 1 - p$$

Step 5. Conclusion: If $|z| < \left|z_{\frac{\alpha}{2}}\right|$, we accept the null hypothesis H_0.
If $|z| > \left|z_{\frac{\alpha}{2}}\right|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Examples
Example 1: In a random sample of 1000 persons from town A, 400 are found to be consumers of wheat. In a sample of 800 from town B, 400 are found to be consumers of wheat. Do these data reveal a significant difference between town A and town B, so far as the proportion of wheat consumers is concerned?
Solution. Given
sample size of town A = $n_1 = 1000$ (≥ 30, large sample)

sample size of town B = $n_2 = 800$ (≥ 30, large sample)
number of consumers of wheat from town A = 400
sample proportion of consumers of wheat in town A, $p_1 = \frac{400}{1000} = 0.4$
number of consumers of wheat from town B = 400
sample proportion of consumers of wheat in town B, $p_2 = \frac{400}{800} = 0.5$
By method of pooling, $p = \frac{n_1p_1+n_2p_2}{n_1+n_2} \Rightarrow p = \frac{1000\times0.4+800\times0.5}{1000+800} \Rightarrow p = \frac{4}{9}$
and $q = 1 - p \Rightarrow q = 1 - \frac{4}{9} \Rightarrow q = \frac{5}{9}$
Let P_1 and P_2 be population proportions of consumers of wheat from towns A, B respectively.
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P_1 = P_2$ i.e., there is no difference
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $P_1 \neq P_2$ i.e., there is a difference (Two-tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$. Critical value of z is $\left|z_{\frac{\alpha}{2}}\right| = 1.96$.
Step 4. Test Statistic:

$$z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \Rightarrow z = \frac{0.4 - 0.5}{\sqrt{\frac{4}{9} \times \frac{5}{9}\left(\frac{1}{1000} + \frac{1}{800}\right)}} \Rightarrow z = -4.242$$

$\therefore |z| = 4.242\ and\ |z| > \left|z_{\frac{\alpha}{2}}\right|$
Step 5. Conclusion: $\because |z| > \left|z_{\frac{\alpha}{2}}\right|$, we reject the null hypothesis H_0 at 5% level of significance.
We accept the alternative hypothesis H_1 i.e., there is significant difference between town A and town B, as far as the proportion of wheat consumers is concerned.

Example 2: Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, at 5% level of significance.
Solution. Given
sample size of men = $n_1 = 400$ (≥ 30, large sample)
sample size of women = $n_2 = 600$ (≥ 30, large sample)
number of men in favour of proposal = 200
sample proportion of men in favour of proposal, $p_1 = \frac{200}{400} = 0.5$
number of women in favour of proposal = 325
sample proportion of women in favour of proposal, $p_2 = \frac{325}{600} = 0.541$
By method of pooling, $p = \frac{n_1p_1+n_2p_2}{n_1+n_2} \Rightarrow p = \frac{400\times0.5+600\times0.541}{400+600} \Rightarrow p = 0.525$
and $q = 1 - p \Rightarrow q = 1 - 0.525 \Rightarrow q = 0.475$
Let P_1 and P_2 be population proportions of men and women respectively.
Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $P_1 = P_2$ i.e., there is no significant difference between proportion of all men and all women.
Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $P_1 \neq P_2$ (Two-tailed test)
Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$. Critical value of z is $\left|z_{\frac{\alpha}{2}}\right| = 1.96$.
Step 4. Test Statistic:

$$z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \Rightarrow z = \frac{0.5 - 0.541}{\sqrt{(0.525) \times (0.475)\left(\frac{1}{400} + \frac{1}{600}\right)}} \Rightarrow z = -1.28$$

$\therefore |z| = 1.28\ and\ |z| < \left|z_{\frac{\alpha}{2}}\right|$

Step 5. Conclusion: $\because |z| < \left|z_{\frac{\alpha}{2}}\right|$, we accept the null hypothesis H_0 at 5% level of significance and conclude that there is no significant difference of opinion between men and women as far as proposal of flyover is concerned.

Test of significance for small samples (Small samples, $n < 30$)
Student's t-test for single mean
Step 1. Null Hypothesis (H_0): $\mu = \mu_0$.
Step 2. Alternative Hypothesis (H_1): (i) $\mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) (Two-tailed test)
Or (ii) $\mu > \mu_0$ (Right tailed test)
Or (iii) $\mu < \mu_0$ (Left tailed test)
Step 3. Level of Significance (α): Set the level of significance α.
Critical value of t is t_α (one-tailed) or $t_{\frac{\alpha}{2}}$ (two-tailed) with degree of freedom $\nu = n - 1$.
Step 4. Test Statistic:

(i) $t = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n-1}}}$, if sample standard deviation(s) is given.

(ii) $t = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n}}}$, if sample standard deviation not given directly. Here $S^2 = \frac{1}{n-1}\sum(x_i - \overline{x})^2$

Step 5. Conclusion: If $|t_{cal}| < |t|$, we accept the null hypothesis H_0.
If $|t_{cal}| > |t|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Example 1: A random sample of six steel beams has a mean compressive strength of 58391 p.s.i. (pounds per square inch) with a standard deviation of 648 p.s.i. Use this information and the level of significance $\alpha = 0.05$ to test whether the true average compressive strength of the steal from which this sample came is 58000 p.s.i. Assume normality.
Solution. Given
sample size $n = 6$ (< 30, small sample),
sample mean $\overline{x} = 58392$,
sample standard deviation $s = 648$,
degrees of freedom $\nu = n - 1 \Rightarrow \nu = 6 - 1 \Rightarrow \nu = 5$,
True(population) average compressive strength of the steel $\mu = 58000$
Step 1. Null Hypothesis (H_0): $\mu = 58000$.
Step 2. Alternative Hypothesis (H_1): $\mu \neq 58000$ (Two-tailed test)
Step 3. Level of Significance (α): $\alpha = 0.05$.
Step 4. Test Statistic: $t = \frac{\overline{x}-\mu}{\frac{s}{\sqrt{n-1}}} \Rightarrow t = \frac{58392-58000}{\frac{648}{\sqrt{5}}} = 1.353$
For $\nu = 5, t_{\frac{\alpha}{2}} = t_{0.025} = 2.571$ (table)
Step 5. Conclusion: Since $|t| < \left|t_{\frac{\alpha}{2}}\right|$, we accept the null hypothesis and conclude that the average compressive strength of the steel beam is 58000 p.s.i.

Example 2: A random sample of 10 boys had the following I.Q.s: 70, 120, 110, 101, 88, 83, 95, 98, 107, and 100.
Solution. Given
sample size $n = 10$ (< 30, small sample),
sample = $\{x_i\} = \{70, 120, 110, 101, 88, 83, 95, 98, 107, 100\}$

sample mean $\overline{x}$ is given by $\overline{x} = \frac{\sum x_i}{n} \Rightarrow \overline{x} = \frac{70+120+110+101+88+83+95+98+107+100}{10} = 97.2$

sample variance s^2 is given by $S^2 = \frac{1}{n-1}\sum(x_i - \overline{x})^2$

x_i	$x_i - \overline{x}$	$(x_i - \overline{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
$\sum x_i = 972$		$\sum(x_i - \overline{x})^2$ =1833.60

$S^2 = \frac{1}{9}(1833.60) = 203.73$

Sample standard deviation is $S = \sqrt{203.73} = 14.27$

Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $\mu = 100$.

Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: $\mu \neq 100$ (Two-tailed test)

Step 3. Level of Significance $(\boldsymbol{\alpha})$: $\alpha = 0.05$.

Step 4. Test Statistic: $t = \frac{\overline{x}-\mu}{\frac{S}{\sqrt{n}}}$ (sample S.D. is not given directly)

$$\Rightarrow t = \frac{97.2 - 100}{\frac{14.27}{\sqrt{10}}} = -0.62$$

For $\nu = n - 1 = 10 - 1 = 9, t_{\frac{\alpha}{2}} = t_{0.025} = 2.262$ (table)

Step 5. Conclusion: Since $|t| < \left|t_{\frac{\alpha}{2}}\right|$, we accept the null hypothesis and conclude that the population mean I.Q. is 100.

Student's t-test for difference of means

Let $\overline{x}$ and $\overline{y}$ be the means of two independent samples of sizes $n_1(< 30)$ and $n_2(< 30)$ drawn from two normal populations having means μ_1 and μ_2. To test whether the two population means are equal (i.e., to test whether the difference $\mu_1 - \mu_2$ is significant).

Step 1. Null Hypothesis $(\boldsymbol{H_0})$: $\mu_1 = \mu_2$.

Step 2. Alternative Hypothesis $(\boldsymbol{H_1})$: (i) $\mu_1 \neq \mu_2$ (i.e., $\mu_1 > \mu_2 or \mu_1 < \mu_2$) (Two-tailed test)
Or (ii) $\mu_1 > \mu_2$ (Right tailed test)
Or (iii) $\mu_1 < \mu_2$ (Left tailed test)

Step 3. Level of Significance $(\boldsymbol{\alpha})$: Set the level of significance α. Critical value of t is t_α (one-tailed) or $t_{\frac{\alpha}{2}}$ (two-tailed) with degrees of freedom $\nu = n_1 + n_2 - 2$.

Step 4. Test Statistic: $t = \frac{\overline{x}-\overline{y}}{S\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$ follows t-distribution with $\nu = n_1 + n_2 - 2$ d.f.

where $S^2 = \frac{1}{n_1+n_2-2}[\sum(x_i - \overline{x})^2 + \sum(y_i - \overline{y})^2]$ or $S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2-2}$

Step 5. Conclusion: If $|t| < |t_\alpha|$, we accept the null hypothesis H_0.

If $|t| > |t_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Example 1: To examine the hypothesis that the husbands are more intelligent than the wives, an investigator took a sample of 10 couples and administered them a test which measures the I.Q.. The results are as follows:

Husbands	117	105	97	105	123	109	86	78	103	107
Wives	106	98	87	104	116	95	90	69	108	85

Test the hypothesis with a reasonable test at the level of significance of 0.05.

Solution.

We have

sample size $n_1 = 10$ (< 30, small sample)

sample size $n_2 = 10$ (< 30, small sample)

sample mean $\overline{x} = \frac{117+105+97+105+123+109+86+78+103+107}{10} = 103$,

sample mean $\overline{y} = \frac{106+98+87+104+116+95+90+69+108+85}{10} = 95.8$

x	$x - \overline{x}$	$(x - \overline{x})^2$	y	$y - \overline{y}$	$(y - \overline{y})^2$
117	14	196	106	10.2	104.04
105	2	4	98	2.2	4.84
97	-6	36	87	-8.8	77.44
105	2	4	104	8.2	67.24
123	20	400	116	20.2	408.04
109	6	36	95	-0.8	0.64
86	-17	289	90	-5.8	33.64
78	-25	625	69	-26.8	718.24
103	0	0	108	12.2	148.84
107	4	16	85	-10.8	116.64
		$\sum(x - \overline{x})^2$ =1606			$\sum(y - \overline{y})^2$ =1679.6

$$S^2 = \frac{1}{n_1 + n_2 - 2}\left[\sum(x_i - \overline{x})^2 + \sum(y_i - \overline{y})^2\right]$$

$$= \frac{1}{18}[1606 + 1679.6] = 182.53$$

$S = \sqrt{182.53} = 13.51$

Step 1. Null Hypothesis (H_0): $\mu_1 = \mu_2$ (no difference in I.Q.)

Step 2. Alternative Hypothesis (H_1): $\mu_1 > \mu_2$ (Right tailed test)

Step 3. Level of Significance (α): $\alpha = 0.05$ (95% confidence)

Critical value of t is $|t_\alpha| = 1.734$ with d.f $\nu = 10 + 10 - 2 = 18$.

Step 4. Test Statistic: $t = \frac{\overline{x} - \overline{y}}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{103 - 95.8}{13.51\sqrt{\frac{1}{10} + \frac{1}{10}}} = 1.19168$

$\therefore\ |t| < |t_\alpha|$

Step 5. Conclusion: Since $|t| < |t_\alpha|$, we accept the null hypothesis H_0 i.e., there is no difference in I.Q.s of husbands and wives.

Snedecor's F-test of significance

When testing the significance of the difference of the means of two samples, we assumed that the two samples came from the same population or from populations with equal variances. If the variances of the populations are not equal, a significant difference in the means may arise. Hence, before we apply the t-test for the significance of the difference of two means, we have to test for the equality of population variances using F-test of significance.

Test for equality of two population variances

Let two independent random samples of sizes n_1 and n_2 be drawn from two normal populations.

To test the hypothesis that the two population variances σ_1^2 and σ_2^2 are equal.

Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

Step 3. Level of Significance (α): Set the level of significance α.

$F_\alpha(v_1, v_2)\ or\ F_{cal}$ is the value of F with $v_1 = n_1 - 1\ and\ v_2 = n_2 - 1$ degrees of freedom such that the area under the F-distribution to the right of F_α is α.

Step 4. Test Statistic: $F = \frac{S_1^2}{S_2^2}\ or\ \frac{S_2^2}{S_1^2}\ or\ \frac{greater\ variance}{smaller\ variance}$

S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2

$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{1}{n_1 - 1}[\sum(x_i - \overline{x})^2]$ and $S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{1}{n_2 - 1}[\sum(y_i - \overline{y})^2]$

Step 5. Conclusion: If $F < F_\alpha$, we accept the null hypothesis H_0.

If $F > F_\alpha$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Examples

Example 1: In one sample of 8 observations from a normal population, the sum of the squares of deviations of the sample values from the sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5% level whether the populations have the same variance.

Solution. Given

sample size $n_1 = 8$ (<30, small sample) and

sum of squares of deviations of the sample values from the sample mean = 84.4 i.e., $\sum(x_i - \overline{x})^2 = 84.4$,

sample size $n_2 = 10$ (<30, small sample) and

sum of squares of deviations of the sample values from the sample mean = 102.6 i.e., $\sum(y_i - \overline{y})^2 = 102.6$.

Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

If S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2, then

$S_1^2 = \frac{1}{n_1 - 1}[\sum(x_i - \overline{x})^2] = \frac{1}{7}(84.4) = 12.057$ and

$S_2^2 = \frac{1}{n_2 - 1}\left[\sum(y_i - \overline{y})^2\right] = \frac{1}{9}(102.6) = 11.4$

Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

Step 3. Level of Significance (α): $\alpha = 0.05$.

$F_\alpha(v_1, v_2) = F_{0.05}(7,9) = 3.29$ with $v_1 = n_1 - 1 = 8 - 1 = 7 and\ v_2 = n_2 - 1 = 10 - 1 = 9$ degrees of freedom

Step 4. Test Statistic: $F = \frac{S_1^2}{S_2^2}$ since $S_1^2 > S_2^2$

$$F = \frac{S_1^2}{S_2^2} = \frac{12.057}{11.4} = 1.057$$

$F < F_{cal}$

Step 5. Conclusion: If $F < F_{\alpha}$, we accept the null hypothesis H_0 and conclude that the populations have the same variance.

Example 2: Time taken by the workers in performing a job by method I and method II is given below:

Method I	20	16	26	27	23	22	---
Method II	27	33	42	35	32	34	38

Do the date show that the variances of time distribution from population from which these samples are drawn do not differ significantly?

Solution. Given

sample size $n_1 = 6$ (<30, small sample) and

sample mean $\overline{x} = \frac{\sum x}{n_1} = \frac{20+16+26+27+23+22}{6} = 22.3$

sample size $n_2 = 7$ (<30, small sample) and

sample mean $\overline{y} = \frac{\sum y}{n_2} = \frac{27+33+42+35+32+34+38}{7} = 34.4$

x	$x - \overline{x}$	$(x - \overline{x})^2$	y	$y - \overline{y}$	$(y - \overline{y})^2$
20	-2.3	5.29	27	-7.4	54.76
16	-6.3	39.69	33	-1.4	1.96
26	3.7	13.69	42	7.6	57.76
27	4.7	22.09	35	0.6	0.36
23	0.7	0.49	32	-2.4	5.76
22	-0.3	0.09	34	-0.4	0.16
			38	3.6	12.96
		$\sum(x - \overline{x})^2$ =81.34			$\sum(y - \overline{y})^2$ =133.72

Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

If S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2, then

$S_1^2 = \frac{1}{n_1 - 1}[\sum(x_i - \overline{x})^2] = \frac{1}{5}(81.34) = 16.26$ and

$S_2^2 = \frac{1}{n_2 - 1}\left[\sum(y_i - \overline{y})^2\right] = \frac{1}{6}(133.72) = 22.29$

Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value is

$F_{\alpha}(v_1, v_2) = F_{0.05}(5,6) = 4.39$ with $v_1 = n_1 - 1 = 6 - 1 = 5$ *and* $v_2 = n_2 - 1 = 7 - 1 = 6$ degrees of freedom

Step 4. Test Statistic: $F = \frac{S_2^2}{S_1^2}$ since $S_2^2 > S_1^2$

$$F = \frac{S_2^2}{S_1^2} = \frac{22.29}{16.26} = 1.37$$

$F < F_{\alpha}$

Step 5. Conclusion: If $F < F_{\alpha}$, we accept the null hypothesis H_0 and conclude that the populations have the same variance.

2. **The nicotine contents in milligrams in two samples of tobacco were found to be as follows.**

Samples A	**24**	**27**	**26**	**21**	**25**	
Samples B	**27**	**30**	**28**	**31**	**22**	**36**

Can it be said that the two samples have came from the same normal population?

Solution:- (i) F-test for checking variances we have to do

$n_1 = 5, n_2 = 6$ (ii) t-test for checking means

If both means & variances are same → same normal population

$$\bar{x} = \frac{\sum x_i}{n_1} = \frac{24+27+26+21+25}{5} = 24.6$$

$$\bar{y} = \frac{\sum y_i}{n_2} = \frac{27+30+28+31+22+36}{6} = 29$$

x	$x-\bar{x}$	$(x-\bar{x})^2$	y	$y-\bar{y}$	$(y-\bar{y})^2$
24	-0.6	0.36	27	-2	4
27	2.4	5.76	30	1	1
26	1.4	1.96	28	-1	1
21	-3.6	12.96	31	2	4
25	0.4	0.16	22	-7	49
			36	7	49
123		21.2	174		108

$\bar{x} = 24.6$

$\bar{y} = 29$

From the table, $\sum (x_i - \bar{x})^2 = 21.2$

and $\sum (y_i - \bar{y})^2 = 108$.

i) F-test to compare the variances :-

$$s_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1}$$

$= \frac{21.2}{4} = 5.3$

$\therefore S_1^2 = 5.3$

$S_2^2 = \frac{\Sigma(y_i - \bar{y})^2}{n_2 - 1}$

$= \frac{108}{5} = 21.6$

$S_1^2 = 5.3, \quad S_2^2 = 21.6$

$H_0: S_1^2 = S_2^2$ i.e., Two sample variances are from same population

$H_1: S_1^2 \neq S_2^2$

i.e., two sample variances are not from same population

$F\ test = \frac{S_2^2}{S_1^2} \quad [\because S_2^2 > S_1^2]$

$= \frac{21.6}{5.3} = 4.075$

n_1	n_2
5	6
$v_1 = 4$	$v_2 = 5$

(4, 5) d.f 5% level of significance

$F_{tab} = 5.19$

$|F_{cal}| = 4.075$.

$|F_{cal}| < F_{tab}$ $\therefore H_0$ is accepted. $\therefore$ Two samples variances are from same population.

(ii) t-test for comparing means:-

$S^2 = \frac{\Sigma(x_i - \bar{x})^2 + \Sigma(y_i - \bar{y})^2}{n_1 + n_2 - 2}$

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

$= \frac{21.2 + 108}{5 + 6 - 2} = 14.35$

$S = \sqrt{14.35} = 3.78$

$t = \frac{\bar{x} - \bar{y}}{S \times \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

$= \frac{24.6 - 29}{3.78 \times \sqrt{\frac{1}{5} + \frac{1}{6}}}$

$= -1.92$

ν (degrees of freedom) $= n_1 + n_2 - 2$

$= 5 + 6 - 2$

$= 9$

$|t_{cal}| = |-1.92| = 1.92$

At 5% level of significance, for two tailed

tabulated t value = 2.262

0.025

9

$|t_{cal}| < t_{tab}$

$\therefore H_0$ is accepted. i.e., The two samples have same means.

$\therefore$ The two samples came from the same normal population.

Chi-Square (χ^2) test

The magnitude of discrepancy between the theory and observation is given by the quantity χ^2. If $\chi = 0$, the observed and expected frequencies completely coincide. As χ^2 increases, the discrepancy between the observed and theoretical frequencies increases.

If $O_i (i = 1,2, \dots, n)$ is a set of observed (experimental) frequencies and $E_i (i = 1,2, \dots, n)$ is the corresponding set of expected (theoretical) frequencies, then χ^2 is defined as $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$ with $(n-1)$ degrees of freedom.

χ^2 is used to test whether differences between observed and expected frequencies are significant or not.

Note: If the data is given in a series of $'n'$ numbers, then degrees of freedom = $n - 1$.

In case of Binomial distribution, d.f. = $n - 1$.

In case of Poisson distribution, d.f. = $n - 2$.

In case of Normal distribution, d.f. = $n - 2$.

Chi-square distribution is an important continuous probability distribution and it is used in both large and small samples. In chi-square tests, χ^2-distribution is mainly used

(i) to test the goodness of fit.
(ii) to test the independence of attributes.
(iii) to test if the population has a specified value of the variance σ^2.

χ^2 –test as a test of goodness of fit

We use this test to decide whether the discrepancy between theory and experiment is significant or not.

Step 1. Null Hypothesis (H_0): There is no significant difference between observed values and the corresponding expected values. Assuming the H_0 is true, the test statistic χ^2 follows Chi-square distribution with $(n-1)$ d.f.

Step 2. Alternative Hypothesis (H_1): There is significant difference between observed values and expected values.

Step 3. Level of Significance (α): Set the level of significance α. Critical value of χ^2 is χ^2_α with degrees of freedom $\nu = n - 1$.

Step 4. Test Statistic: $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$

Step 5. Conclusion: If $\chi^2 < \chi^2_\alpha$, we accept the null hypothesis H_0.
If $\chi^2 > \chi^2_\alpha$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

Examples

Example 1: A die is thrown 264 times with the following results. Show that the die is biased.

Number appeared on the die	1	2	3	4	5	6
Frequency	40	32	28	58	54	52

Solution.

The expected frequency of each of the numbers 1, 2, 3, 4, 5, 6 is $\frac{264}{6} = 44$

Calculation of χ^2

Observed frequencies (O_i)	**Expected frequencies (E_i)**	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
40	44	16	0.3636
32	44	144	3.2727
28	44	256	5.8181
58	44	196	4.4545
54	44	100	2.2727
52	44	64	1.4545
264	**264**		**17.6362**

Step 1. Null Hypothesis (H_0): The die is unbiased

Step 2. Alternative Hypothesis (H_1): The die is biased

Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value is
$\chi^2_\alpha = \chi^2_{0.05} = 11.070$ with $\nu = n - 1 = 6 - 1 = 5$ degrees of freedom

Step 4. Test Statistic: $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} = 17.6362$

$\therefore \chi^2 > \chi^2_\alpha$

Step 5. Conclusion: Since $\chi^2 > \chi^2_\alpha$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1 and conclude that the die is biased.

Example 2: Fit a binomial distribution to the following data and test for its goodness of fit at level of significance 0.05.

Number of heads	0	1	2	3	4	5
Number of tosses (Frequency)	38	144	342	287	164	25

Example 3: Fit a Poisson distribution to the following data and test for its goodness of fit at level of significance 0.05.

x	0	1	2	3	4
f	419	352	154	56	19

Solution.

Mean, $\lambda = \frac{\sum f_i x_i}{\sum f_i} = \frac{419(0)+352(1)+154(2)+56(3)+19(4)}{419+352+154+56+19} = \frac{904}{1000} = 0.904$

Poisson (Theoretical) frequency distribution is given by

$$NP(X = r) = N\frac{e^{-\lambda}\lambda^r}{r!} = 1000\frac{e^{-0.904}(0.904)^r}{r!}$$

x	0	1	2	3	4	
f	419	352	154	56	19	
$NP(r) = N\frac{e^{-\lambda}\lambda^r}{r!}$	404.9 (406.1)	366.1	165.5	49.9	11.3 (12.4)	997.7 (1000)

Calculation of χ^2

Observed frequencies (O_i)	**Expected frequencies (E_i)**	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
419	406.1	166.4	0.4098
352	366.1	198.8	0.5430
154	165.5	132.3	0.7994
56	49.9	37.2	0.7455
19	12.4	43.56	3.5129
			6.0106

Step 1. Null Hypothesis (H_0): Poisson distribution is good fit

Step 2. Alternative Hypothesis (H_1): Poisson distribution is not good fit

Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value is

$\chi^2_\alpha = \chi^2_{0.05} = 7.815$ with $\nu = n - 2 = 5 - 2 = 3$ degrees of freedom

Step 4. Test Statistic: $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} = 6.0106$

$\therefore \chi^2 < \chi^2_\alpha$

Step 5. Conclusion: Since $\chi^2 < \chi^2_\alpha$, we accept the null hypothesis H_0 and conclude that the Poisson distribution is a good fit to the data.

Chi-square test for independence of attributes:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

2×2 Contingency table:-

	C	D	Total
A	a	b	a+b
B	c	d	c+d
Total	a+c	b+d	N = a+b+c+d

(or)

	C	D
A	a	b
B	c	d

	C	D	Total
A	$E(a) = \frac{(a+b)(a+c)}{N}$	$E(b) = \frac{(a+b)(b+d)}{N}$	a+b
B	$E(c) = \frac{(c+d)(a+c)}{N}$	$E(d) = \frac{(c+d)(b+d)}{N}$	c+d
Total	a+c	b+d	N = a+b+c+d

Degrees of freedom $\nu = (r-1)(c-1)$

where r = no. of rows

c = no. of columns

5. Given the following contingency table for hair color and eye color. Find the value of χ^2. Is there good association between the two?

Hair color					
Eye color		Fair	Brown	Black	Total
	Blue	15	5	20	40
	Grey	20	10	20	50
	Brown	25	15	20	60
	Total	60	30	60	150 = N

Solution:

H_0: The hair colour and eye colour are independent.

H_1: The hair colour and eye colour are dependent.

	Fair	Brown	Black	Total
Blue	$\frac{40 \times 60}{150} = 16$	$\frac{40 \times 30}{150} = 8$	$\frac{40 \times 60}{150} = 16$	40
Grey	$\frac{50 \times 60}{150} = 20$	$\frac{50 \times 30}{150} = 10$	$\frac{50 \times 60}{150} = 20$	50
Brown	$\frac{60 \times 60}{150} = 24$	$\frac{60 \times 30}{150} = 12$	$\frac{60 \times 60}{150} = 24$	60
Total	60	30	60	N = 150

O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
15	16	-1	1	0.0625
5	8	-3	9	1.125
20	16	4	16	1
20	20	0	0	0
10	10	0	0	0
20	20	0	0	0
25	24	1	1	0.042
15	12	3	9	0.75
20	24	-4	16	0.666
				3.6458

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

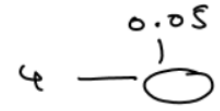
$$= 3.6458$$

r = no. of rows = 3 , c = no. of columns = 3

$$df = (r-1)(c-1) = (3-1)(3-1) = 2 \times 2 = 4.$$

At 5% level of significance for 4 d.f $\chi^2_{tab} = 9.488$

$|\chi^2_{cal}| = 3.6458$, $\chi^2_{tab} = 9.488$



$|\chi^2_{cal}| < \chi^2_{tab}$.

$\therefore H_0$ is accepted.

$\therefore$ The hair colour and eye colour are independent.

$\therefore$ They are not associated.